



Quotation

Project: "Advance Motors - Line 4 OP70"

Customer:

Advance Motors

Customer address:

Fabriksgatan 7, 54150

Skövde - Sweden

Customer reference:

AM0273



Contents

1 Changes applied after quotation review	3
2 Scope of the installation	3
3 Location	3
4 Function, performance and operating conditions	3
5 Time plan	4
6 Price	4
7 Terms of payment	5
8 Changes and supplementary work	5
9 Warranty	5
10 Description	5
9.1 Technical description	5
9.2 Time Plan	7
9.3 Test specifications	7
9.4 Supplier and Customer's commitments	7
9.5 Product specifications	8
9.6 Training	8
9.7 Safety functions	9
9.8 Technical documentation	9
10 CE-marking	9



1 Changes applied after quotation review

- Implementation of a clamp to ensure that objects stop at the correct position and can be picked up with the same placement.
- Change the forklift access. In the pallet introduction there would be a pallet dispenser to guarantee the correct position of the pallet for taking it later. Also light curtains must be added to avoid stopping the production while removing and adding the pallets.
- Facilitate maintenance by placing all sensors on the same side.
- The main HMI panel will be located on the main door for accessing the interior. In the forklift access, there will be an HMI with a light and a button to request access.
- Each conveyor will have their own sensors.
- The operations with the forklift must be done inside the area designated. The last conveyor should be removed to leave space for the forklift operation.
- Moving the security fence closer to the main conveyor to remove the need for an upper door.

2 Scope of the installation

According to the inquiry, the supplier shall deliver and assemble Operation 70 (OP70) of the Line 4 at Advance Motor's factory in Skövde. In OP70, the final operation in the line, products that have been rejected in the previous inspection operation must be taken out from the line and palletized for an optimized removal.

The following quotation includes a turnkey system, including technical documentation, installation of all components and cabinets, controller and HMI panel programming, integration of safety devices, commissioning, manuals and training.

3 Location

The location where the system is being delivered is:

Fabriksgatan 7, 54150

Skövde

Sweden

4 Function, performance and operating conditions

The plant shall meet the following performance and operating characteristics:

- Real capacity must be at least 50 parts per hour.
- Corrected machine capability should be > 1.33 for each refined property.
- Reliability must be at least 90 h
- Availability must be at least 95%



Testing and checking of the installation shall be carried out in accordance with the inquiry specifications, in order to verify its functionality, performance and operating characteristics.

5 Time plan

Relevant milestones and dates that harmonize with your general business terms

- M01 (2024/11/26) - Quotation review
- M02 (2024/12/06) - Design review
- M03 (2024/12/18) - FAT test
- M04 (2025/01/14) - SAT test

6 Price

Item	Description	Cost	Qty	Amount
Engineers	3 weeks	37.500,00 SEK	4	150.000,00 SEK
Instalation/set-up	3 weeks	24.000,00 SEK	4	96.000,00 SEK
Small conveyor		30.000,00 SEK	7	210.000,00 SEK
EuroPallet	800X1200 (mm)	200,00 SEK	10	2.000,00 SEK
Robot	IRB 4400	220.183,00 SEK	1	220.183,00 SEK
Controller	IRC5	30.000,00 SEK	1	30.000,00 SEK
Electrical connections		140.000,00 SEK	1	140.000,00 SEK
Security		300.000,00 SEK	1	300.000,00 SEK
Total				1.008.183,00 SEK

All prices exclude VAT.



7 Terms of payment

Billing takes place after approved SAT, milestone M4. Payment is made 90 days from invoice date.

- 20% of payment will be made at the accepted quotation.
- 10% when validating Technical Documentation.
- 20% at FAT test.
- 50% at approved SAT test.

8 Changes and supplementary work

The following hourly fees shall be used for changes and supplementary work which will affect the quotation amount. These fees represent full hourly compensation to the supplier:

Development/Programming 1300 sek / hour

Installation 700 sek / hour

The above hourly fees are fixed and unchanged until the actual completion date.

9 Warranty

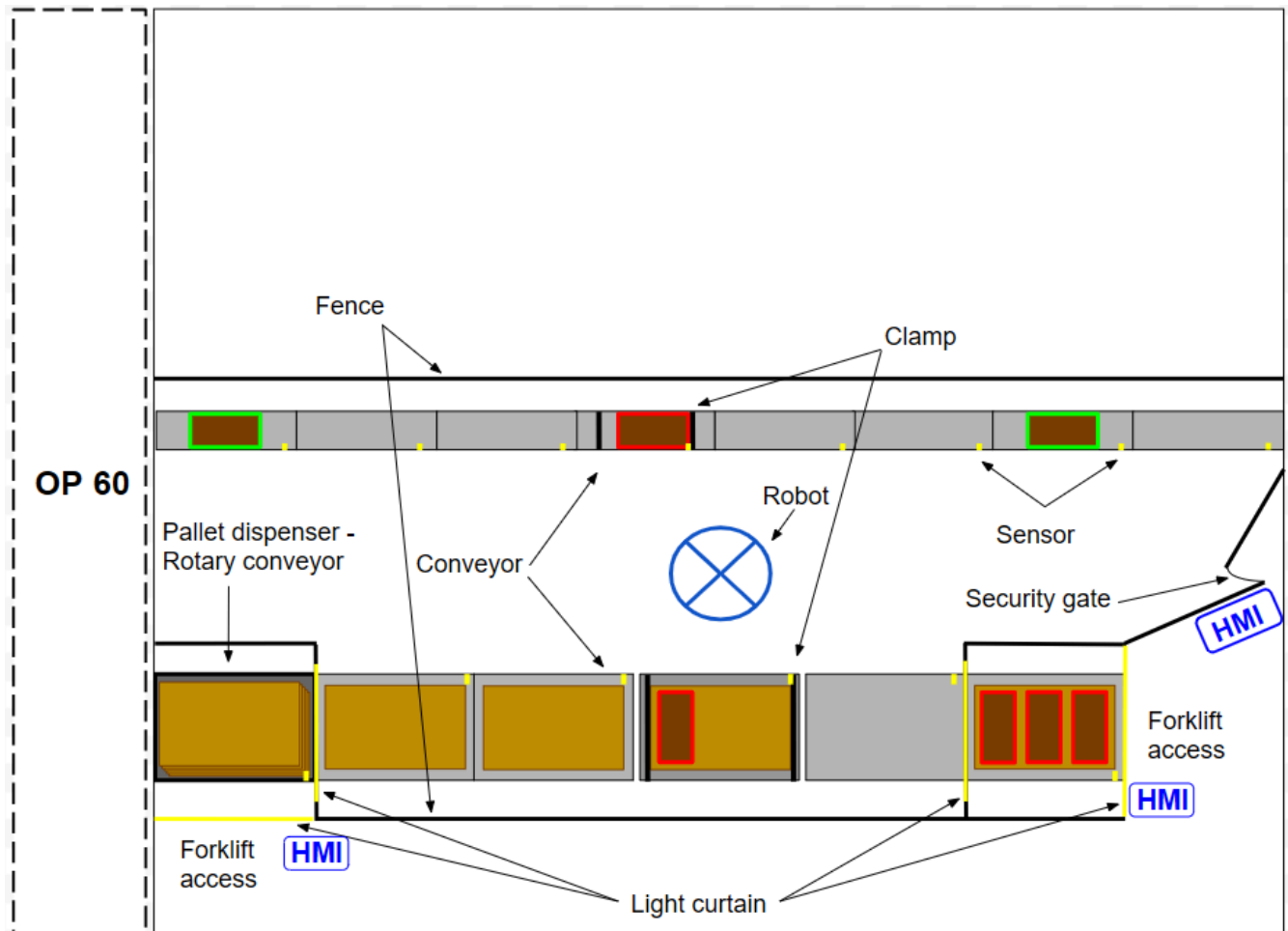
The offer includes 24 months warranty, if maintenance is done according to the delivered description.

The guarantee includes spare parts and shipment, as well as travel, subsistence and accommodation for issues that are not attributed to negligence or external damage.

10 Description

9.1 Technical description

The proposed solution, shown in the figure below, includes 6 conveyors equipped with sensors running at 15m/min. Placed in the direction of production flow: one rotating conveyor will receive a given number of empty pallets and rotate them to face the rest of the conveyor line, this first conveyor will act as a pallet dispenser, forwarding a pallet at a time as they are needed. Following this conveyor are two conveyors that will serve as buffers, a fourth conveyor will keep the pallet being loaded with the rejected products stationary with a clamp, once full, the pallet will be forwarded through another conveyor serving as buffer, to a sixth conveyor, where it will wait for the operator-driven forklift to remove the pallet. The main HMI will be located nearby the security gate.



Fences (2.5m high) are located all around the cell, with the exception of the pallet supplier and removal areas. The access to the first and last conveyors are guarded by double action light curtains, they activate the outer curtains when no pallets are being introduced or removed, and activate the inner ones when the outer ones are disabled. This ensures that there is always a light curtain active between the inside and outside of the production line. As soon as an active light curtain is breached, the emergency stop is executed and the whole process stops. This ensures that no unauthorised elements enter the production area, possibly creating a hazardous situation. Beside both forklift access there will be light indicators and rope-switches to request access to the conveyors.

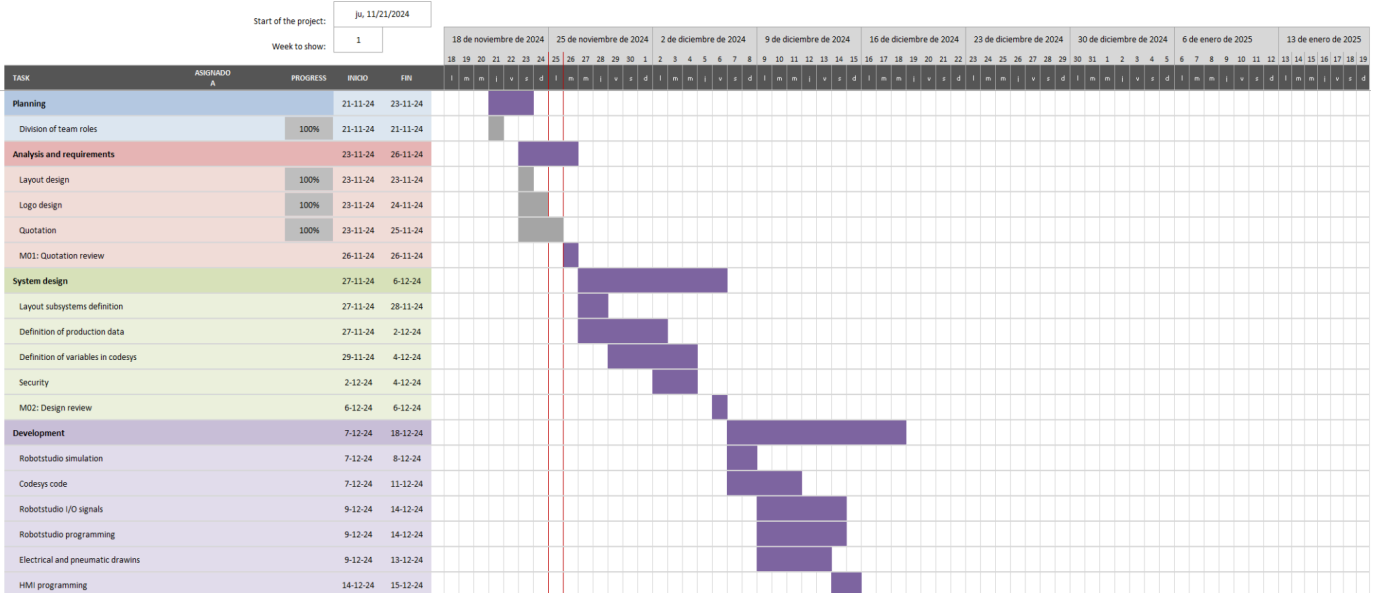
On the main conveyor line arriving from OP60, a clamp system is implemented in the conveyor from which the rejected pieces will be removed, this enables the system to stop the pieces in the exact same spot every time, ensuring repeatability.



9.2 Time Plan

OP70

INTER WHILDS ROBOTICS



OP70

INTER WHILDS ROBOTICS



9.3 Test specifications

InterWilds Robotics will carry out all necessary tests before delivery and after installation. Accepted results for these are a prerequisite for delivery approval or takeover. The tests will be specified in a test protocol agreed with the customer.

9.4 Supplier and Customer's commitments

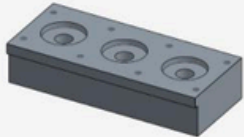
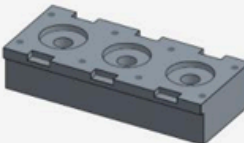
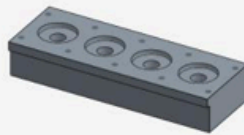
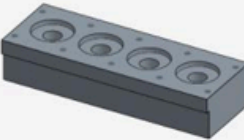
The supplier commits to the delivery of the stipulated robot cell within the time limits established, ensuring its functionality and operations as stated in this document.

The customer commits to provide with the workspace and power management systems needed for the cell to work as intended, as well as the established payment in the time limits given by this document.



9.5 Product specifications

The cell will be able to handle the products included in the inquiry: MB16, MB16T, MB20 and MB24.

Name	Model	Size (l/w/h)	Product Number	Picture
Engine 1.6 Liter	MB16	700x300x150	1001605	
Engine 1.6 Liter Turbo	MB16T	700x300x150	1001610	
Engine 2.0 Liter	MB20	850x300x150	1002004	
Engine 2.4 Liter	MB24	850x300x170	1002407	

9.6 Training

InterWilds Robotics will carry out in-site training for operators and maintenance to ensure the competence for handling the cell.

There will be different sessions to focus on:

Operator training.

Training to Electrical Maintenance, including program review of PLC and Robot

Training for Mechanical Maintenance.



9.7 Safety functions

The cell provided by InterWilds Robotics will include all the necessary safety equipment and safety features specified in the inquiry.

The cell will be equipped with a safety zone through safety gates with their respective panel and emergency stop, an emergency stop in the main panel and at each door to the safety zone, a safety stop that should appear on the main panel display, an exit safety zone and light curtains and fences for the forklift access points.

9.8 Technical documentation

InterWilds Robotics will deliver the following technical documentation:

- Functional description of the operations specific functionalities including all necessary diagrams and flowcharts.
- Electric and Pneumatic drawings, including network diagram and I/O list
- Component list and spare part list.
- User and maintenance manuals.
- Test Protocol.

10 CE-marking

InterWilds Robotics will ensure that the CE marking is performed and documented on the robot cell implemented only, and will not CE mark the plant in its whole.